ATC Interpretability

Graph reachability

We thank the reviewer for this constructive feedback. In response, and in addition to the evidence already in the paper, we ran further probing experiments on the thoughts themselves. They show that ATC's thoughts hold the frontier of a breadth-first search. For the ATC model of Table 1 we applied the model's own output head to every continuous thought (a logit lens) on 5,000 held-out graphs. Fig. R1 shows where that probability falls at each step: thought k puts it on the nodes exactly k hops from the root. Nearly all of the rest is on the two candidates, the target and the distractor, in equal shares of up to about 20% each, so the model does not favour the target before the search reaches it. At the answer step, 98% is on the target. Fig. R2 shows the same on a single randomly chosen graph, step by step. The model carries the whole frontier rather than guessing a path. ATC is supervised on the final answer only, so this structure is learned, not imposed.

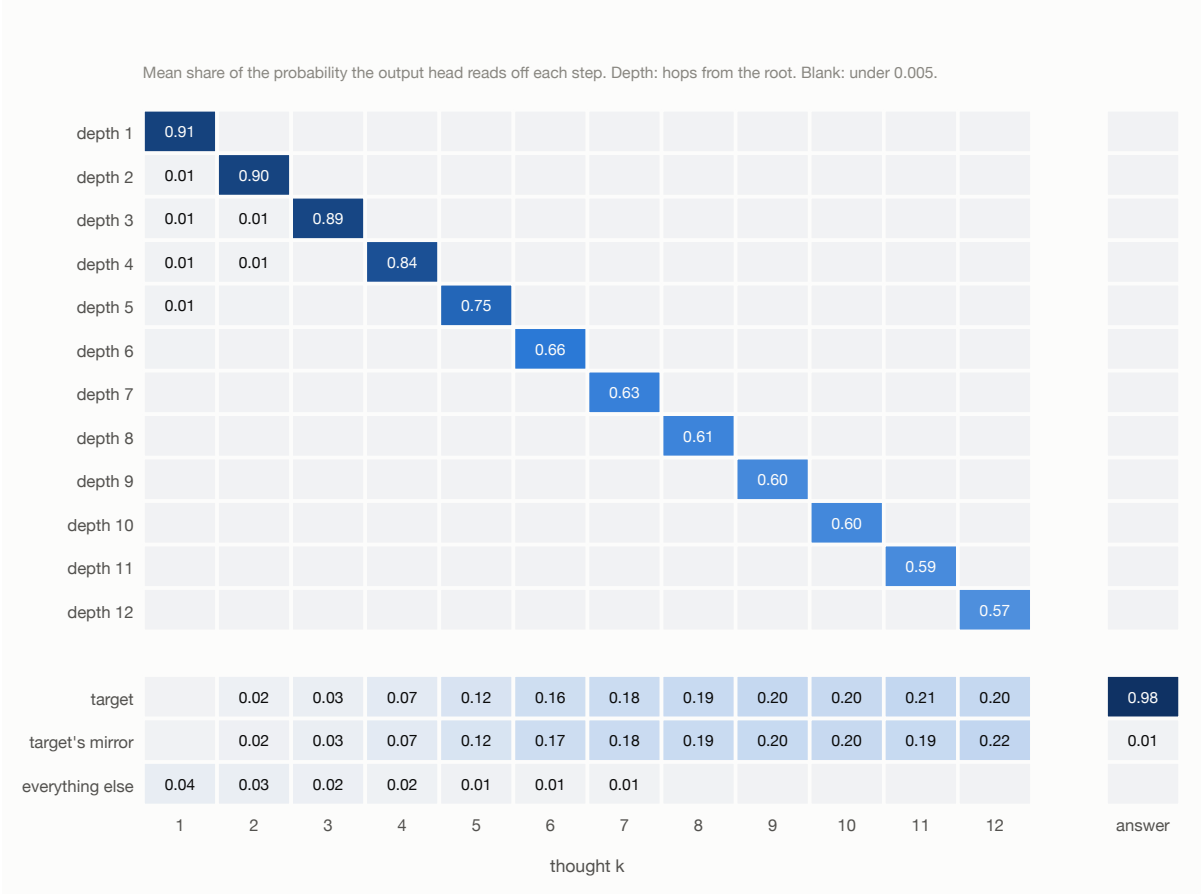
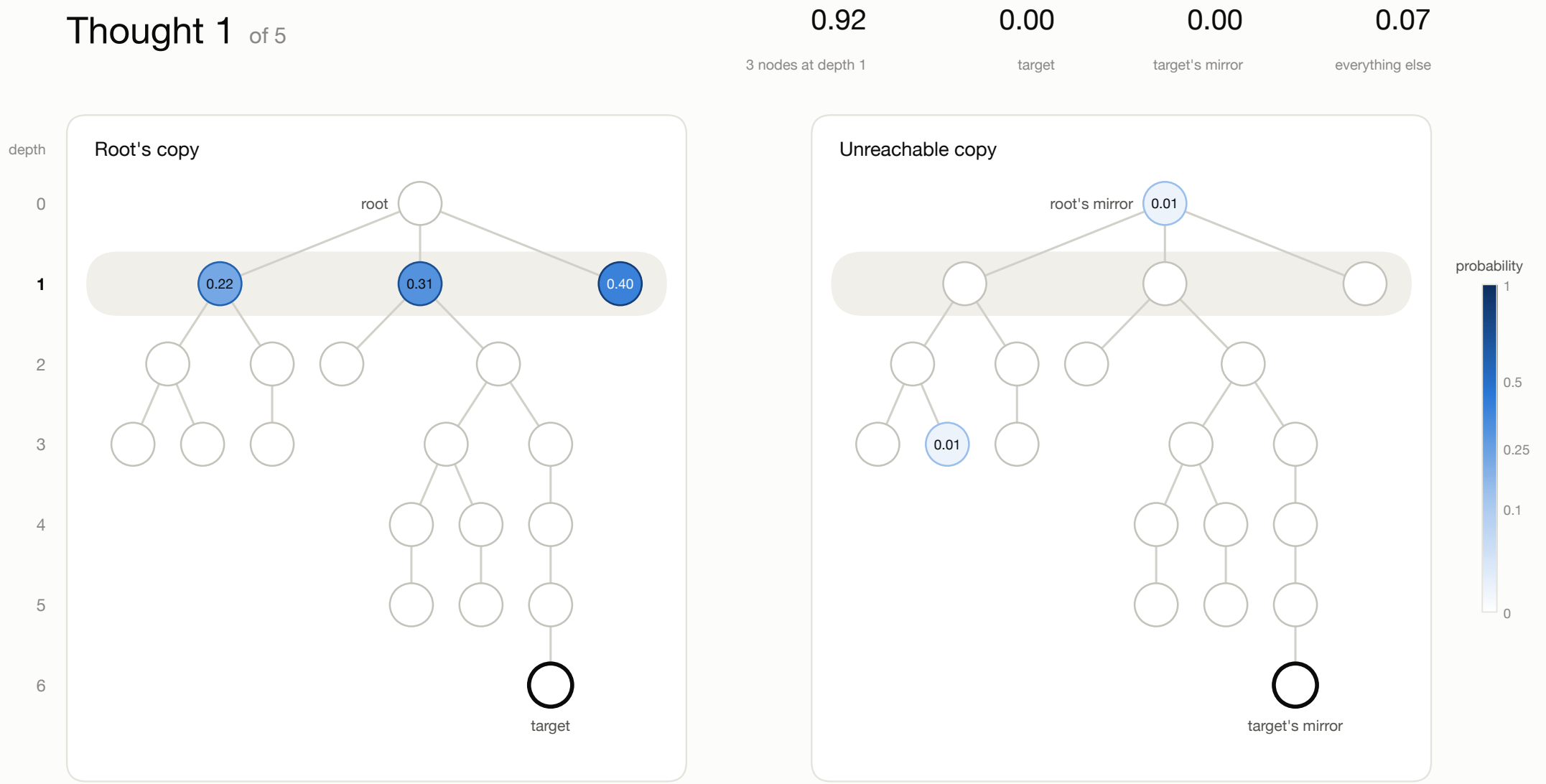
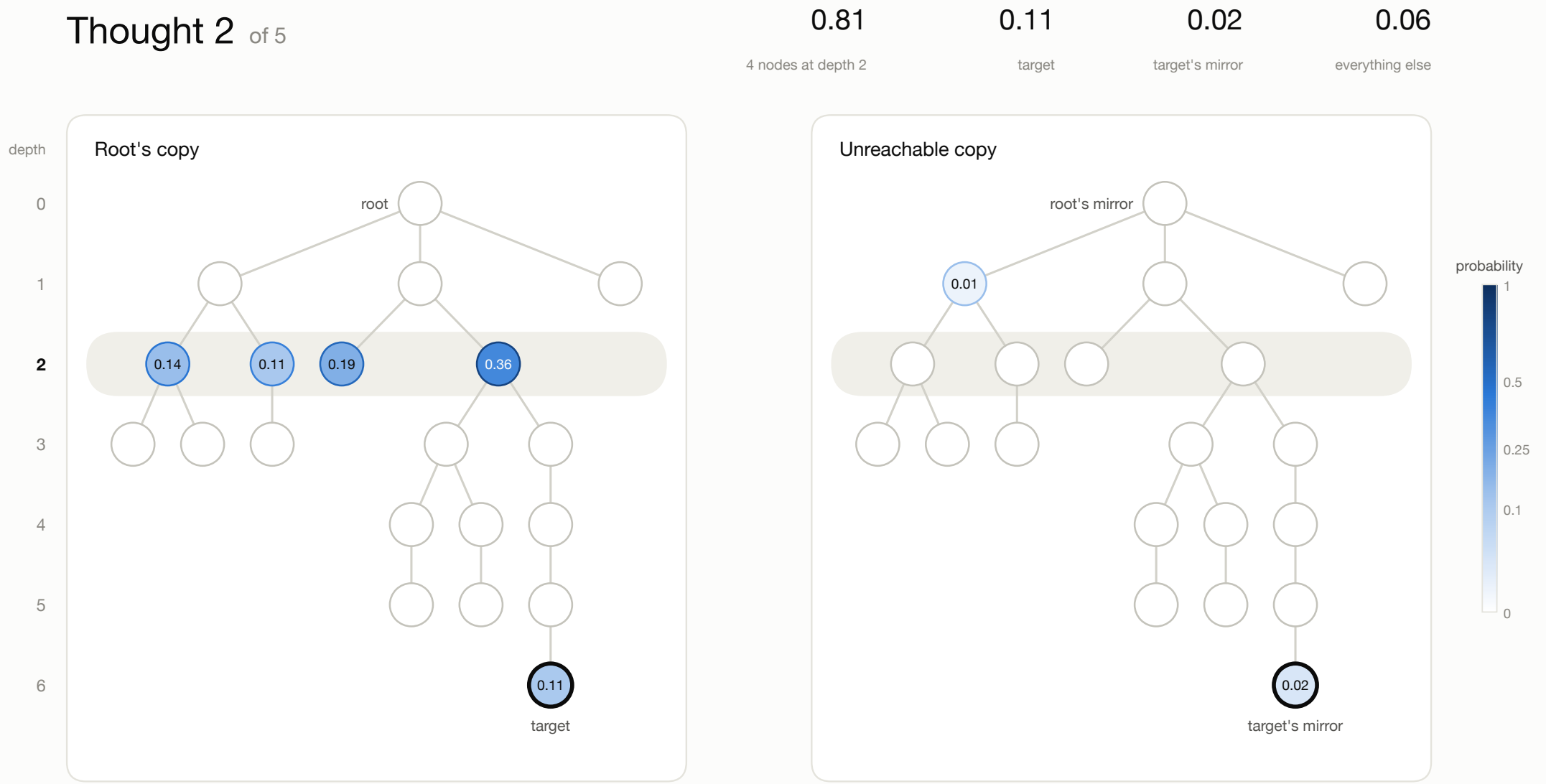


Figure R1



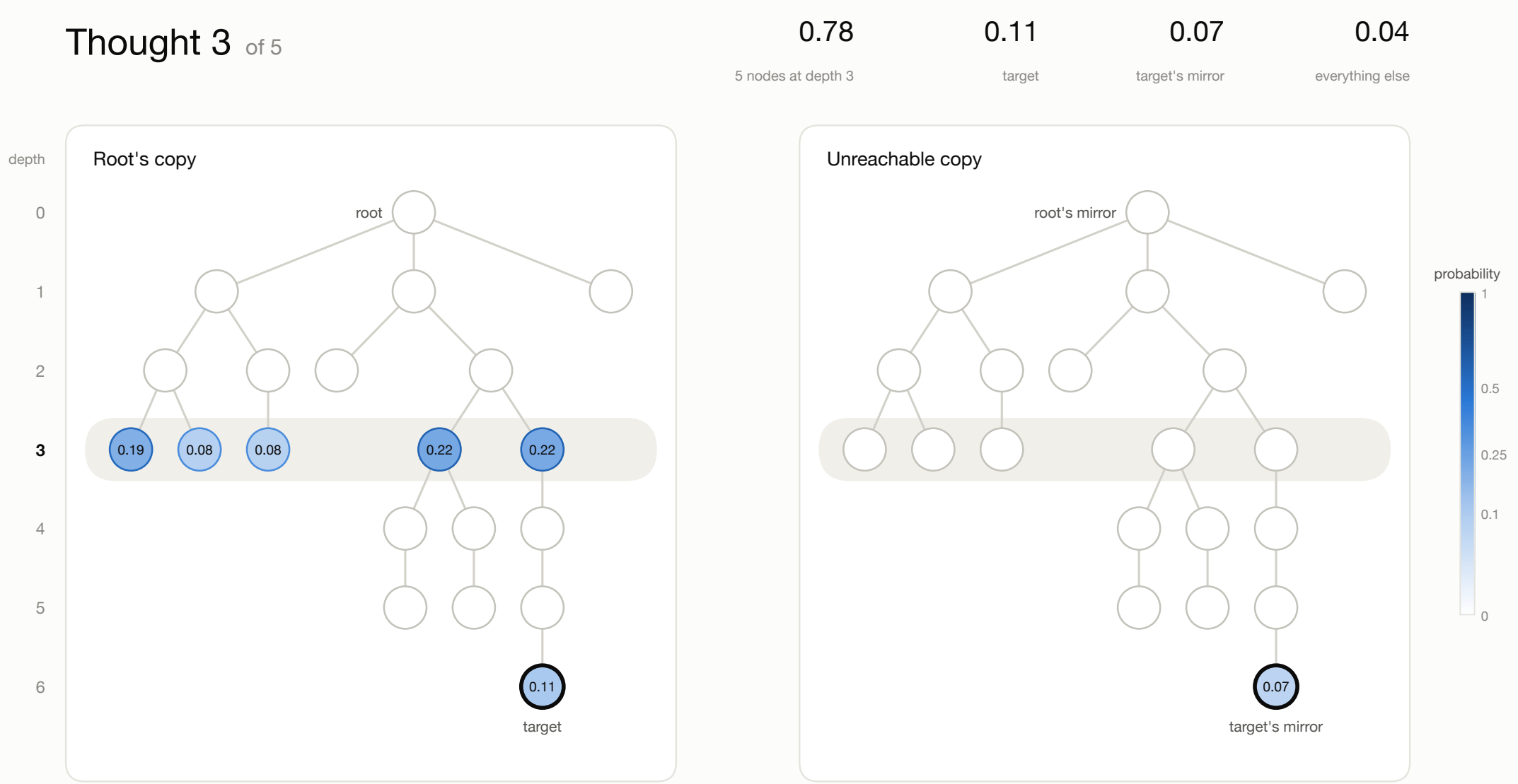
ATC, test graph 0: the target is 6 hops from the root. Each node shows the probability the output head reads off this step (logit lens); blank is under 0.005.

Figure R2



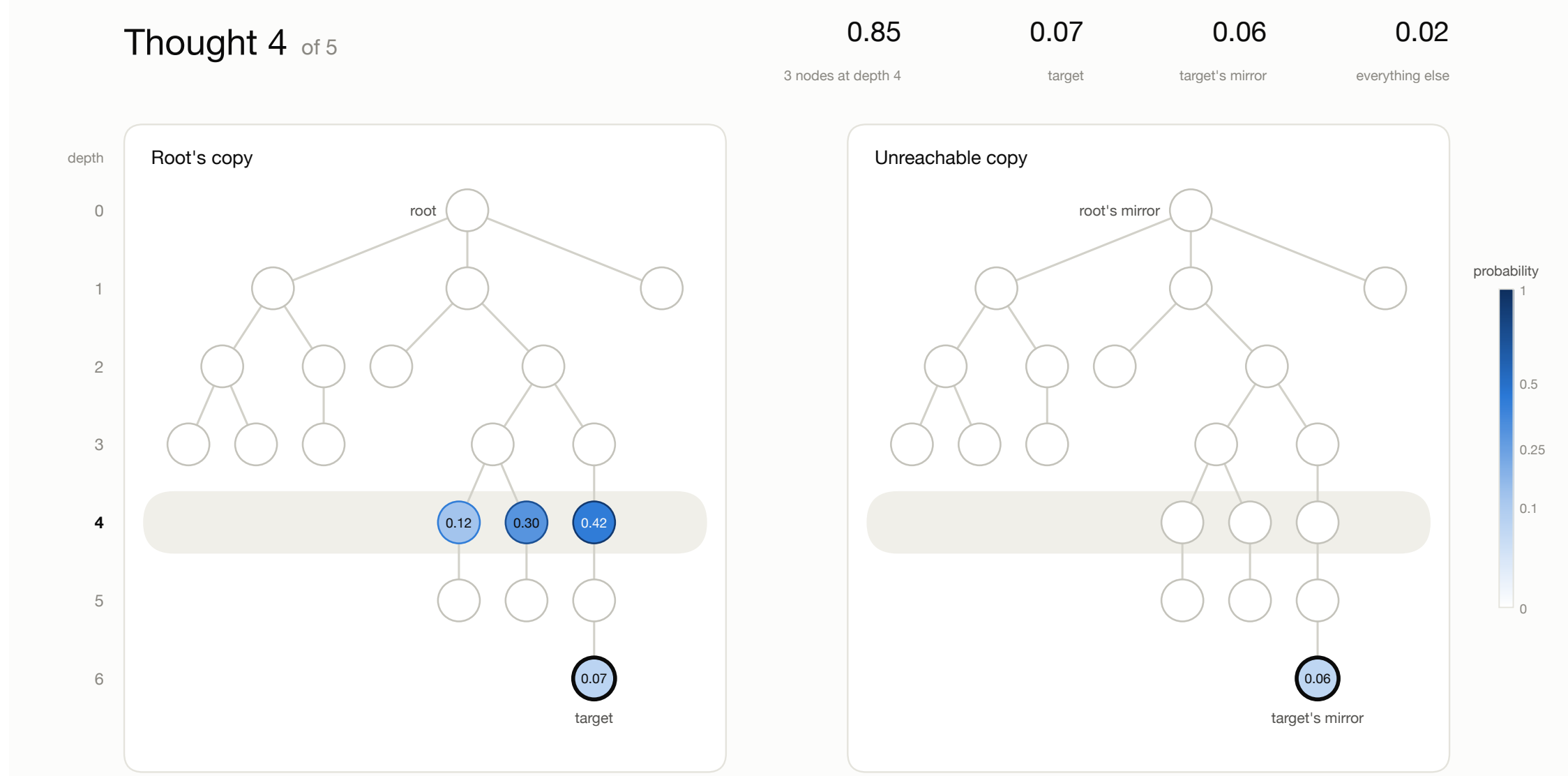
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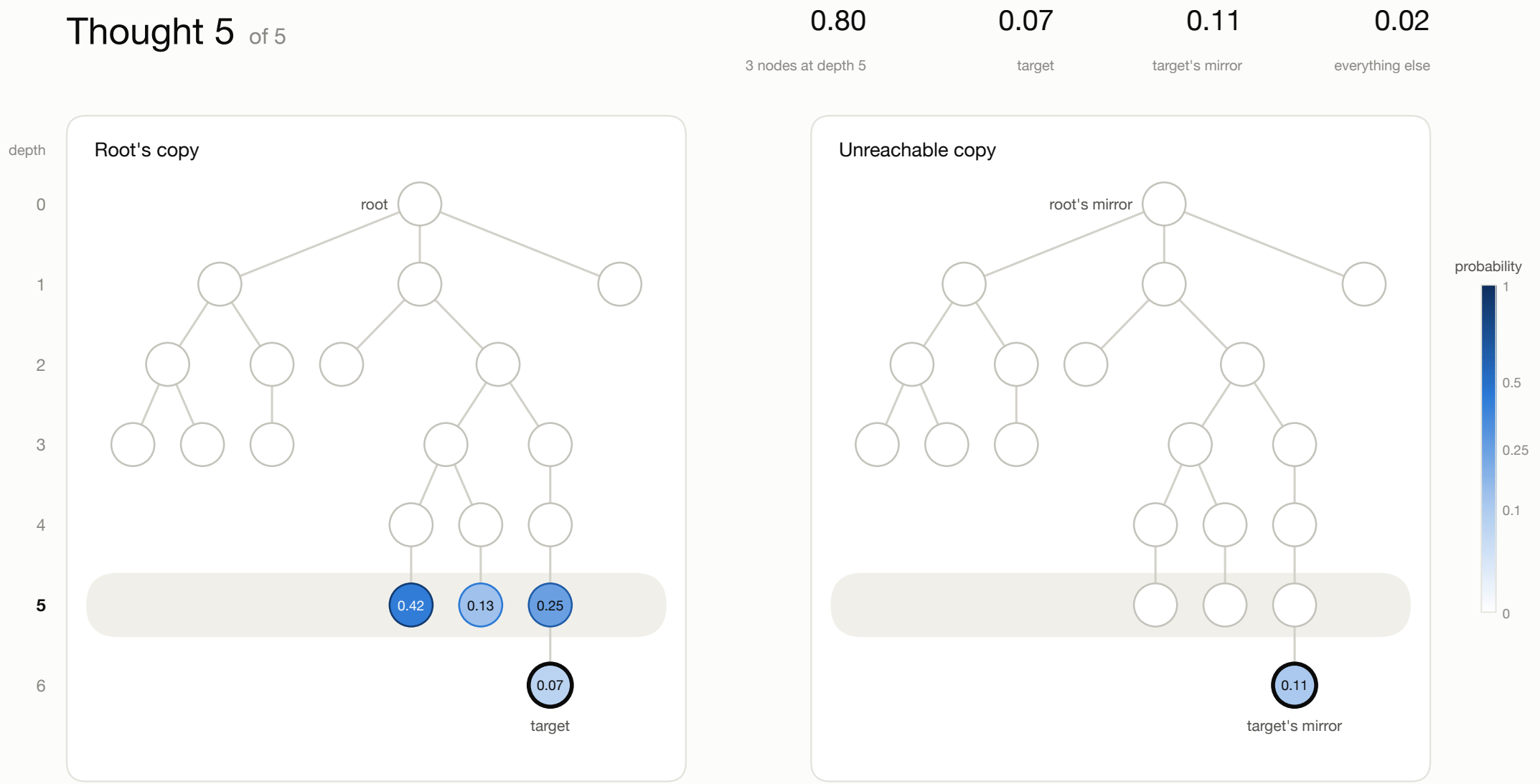
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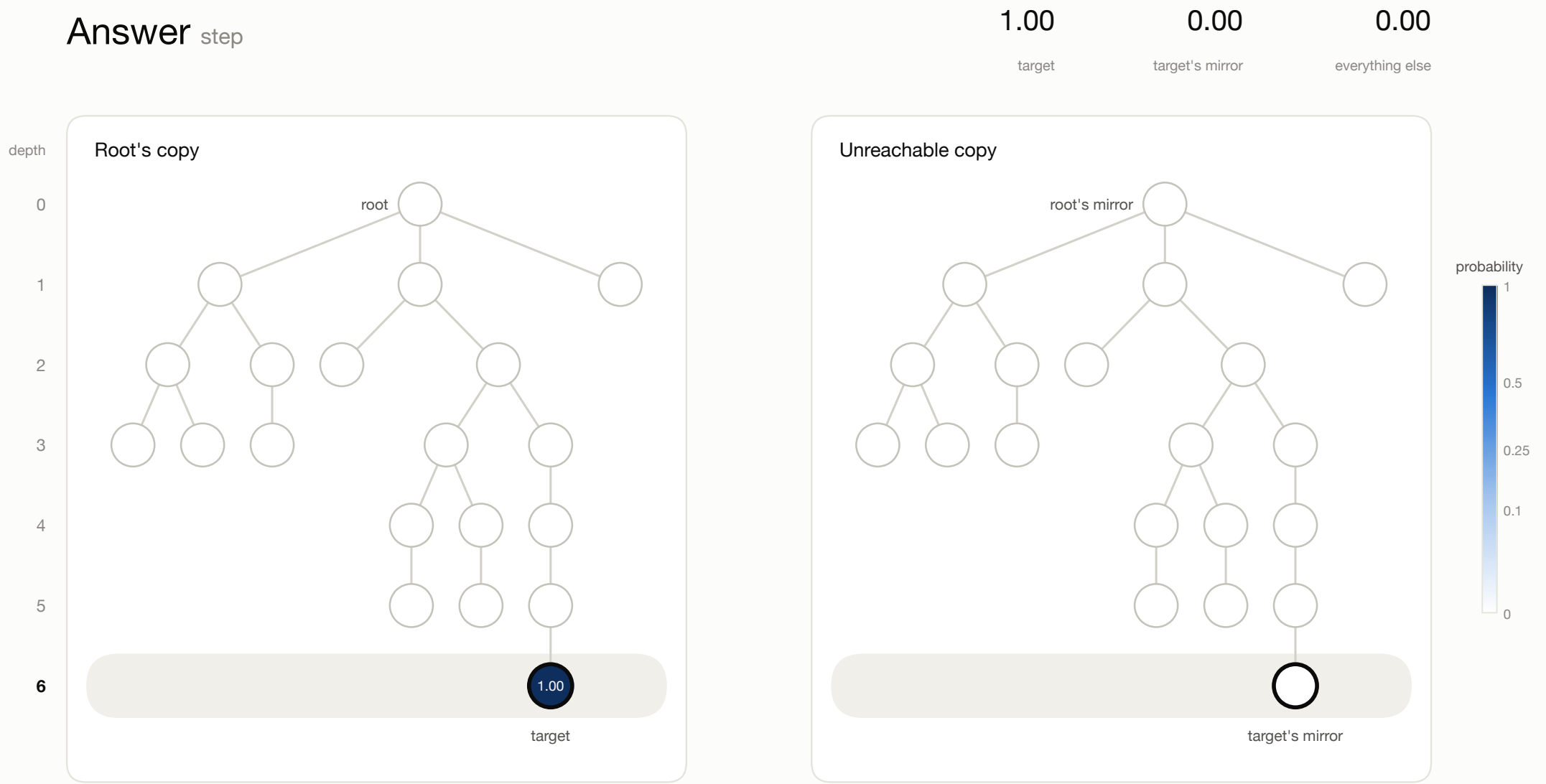
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Figure R2

Addition interpretability.

To test whether the latent thought token (z_i) encodes anything meaningful in addition, we perform the following interpretability experiment. We take the hidden vector $(z_{k+1} = h(z_k))$ and probe it for the true intermediate carry at stage (k). Let (c_k) be the carry at stage (k). For each stage (k), we train an MLP by minimizing the loss $(\text{CrossEntropy}(c_k, \text{MLP}_\theta^{(k)}(h(z_k))))$ to check whether the hidden embedding encodes the carry information at that stage. We also train a probe to map the latent thought (z_i) to the carry at a different position (m not equals i) and measure the prediction accuracy.

In the following figure, we observe that the latent thought at position (i) maps well to the carry at stage (i). However, as expected, the mapping to the carry at a different position performs very poorly, as does the same mapping for a random Transformer.

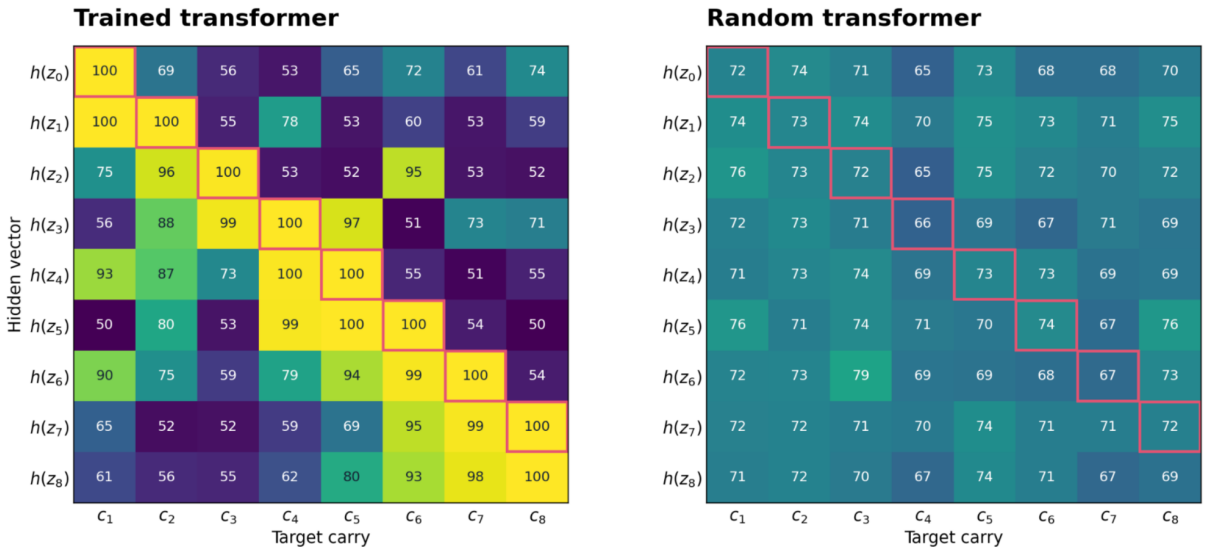


Figure R3

Intermediate supervision

Our curriculum specifies which problems to learn first, but does not specify the intermediate steps for solving them. More generally, useful intermediate states need not be answers to simpler problems at all. Consider learning to check whether brackets in a string match, starting with short strings and gradually increasing their length. Every training label is just "correct" or "incorrect," but the model might learn to remember which brackets still need closing. We never supervise that memory or tell the model how to update it. The curriculum therefore still involves human guidance, but it leaves the intermediate computation, which is much more complicated than the curriculum, to be learned. We will clarify that "no intermediate supervision" means no explicit targets for intermediate reasoning states, rather than no task-specific guidance.